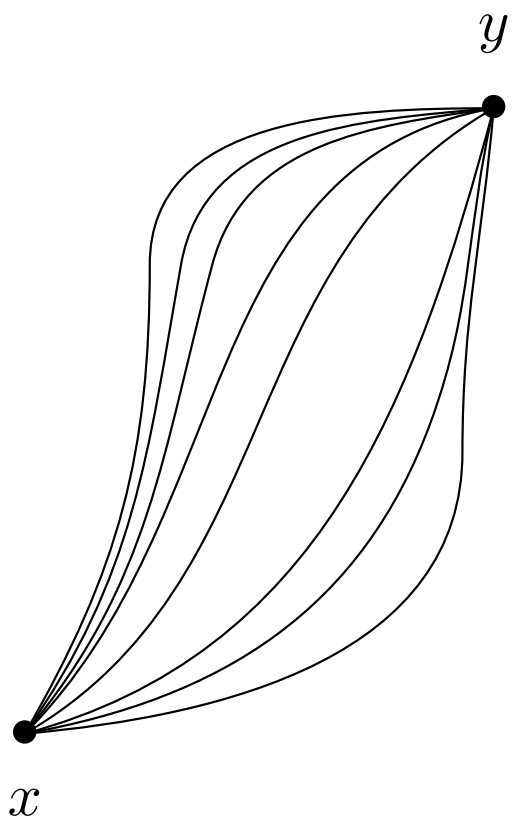
The Seifert-Van Kampen Theorem

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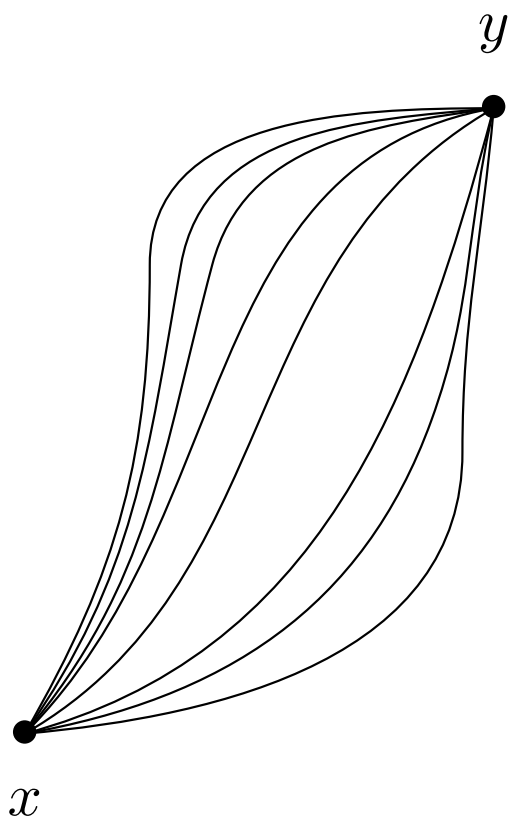
The Fundamental Groupoid

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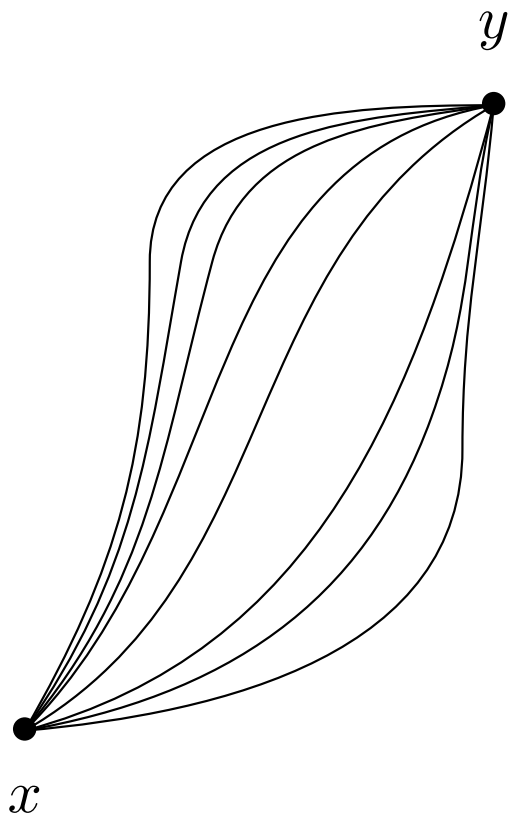
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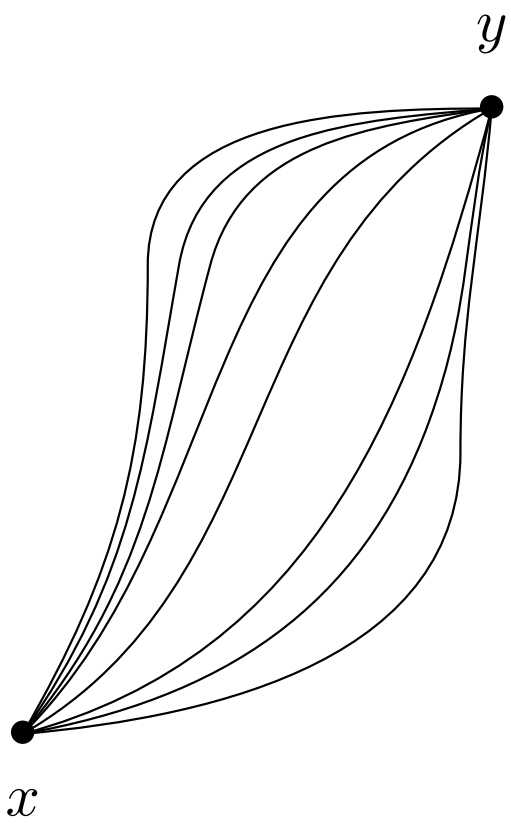
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2. Each way gives what we call a **homotopy of paths**
3. Two paths are equivalent $\implies$ there exists a homotopy between them.
4. **Homotopy classes** are well behaved.



The Fundamental Groupoid

The **fundamental groupoid** is the category (groupoid) $\Pi(X, A)$ where

- The objects are the points of our subspace A of X
- The arrows are the homotopy classes of paths between the points of A

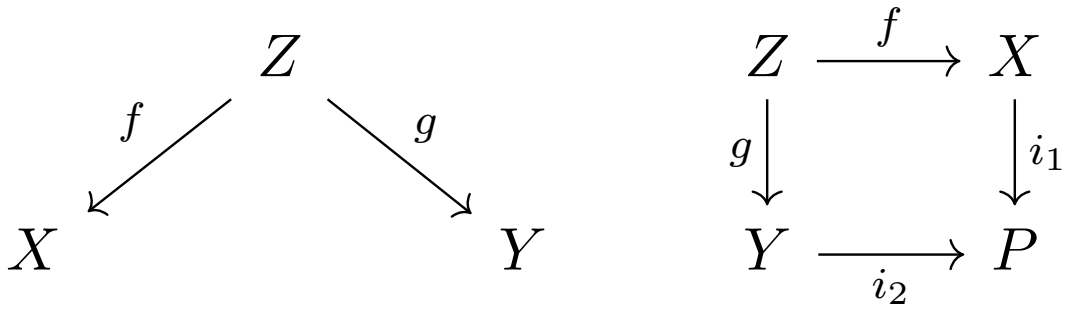
The Fundamental Group

The fundamental group with base point x_0 is formed by the arrows of the groupoid $\Pi(X, \{x_0\})$.

The fundamental group (and groupoid) of a space are homotopy invariant. For instance, homeomorphic spaces have isomorphic fundamental groupoids.

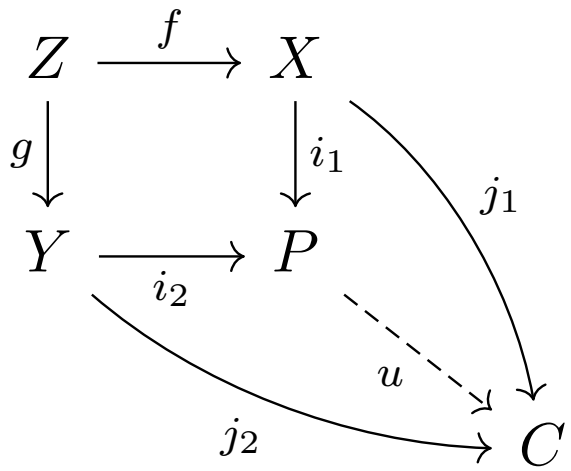
Pushouts in Category Theory

The pushout of a triangle is the "most general" object which can be used to form a commutative square.



Pushouts in Category Theory

This is what is meant by "most general."



The Seifert-Van Kampen Theorem

Suppose X_0, X_1 are open sets with $X = X_0 \cup X_1$ Then

$$\begin{array}{ccc} \Pi(X_0 \cap X_1, A) & \xrightarrow{\Pi(i_0)} & \Pi(X_0, A) \\ \Pi(i_1)\downarrow & & \downarrow\Pi(j_1) \\ \Pi(X_1, A) & \xrightarrow[\Pi(j_0)]{} & \Pi(X, A) \end{array}$$

is a pushout in Grpd.

The Seifert-Van Kampen Theorem

Let's regard $\Pi(-) : \mathsf{Top} \to \mathsf{Grpd}$ as a functor.

It sends topological spaces to their fundamental groupoid, and continuous functions $f : X \to Y$ to the functor

$$\Pi(f) : \Pi(X) \to \Pi(Y), \ [u] \to [fu].$$

In other words, Π **preserves the pushout** of

$$X_0 \leftarrow X_0 \cap X_1 \rightarrow X_1$$

in Top.

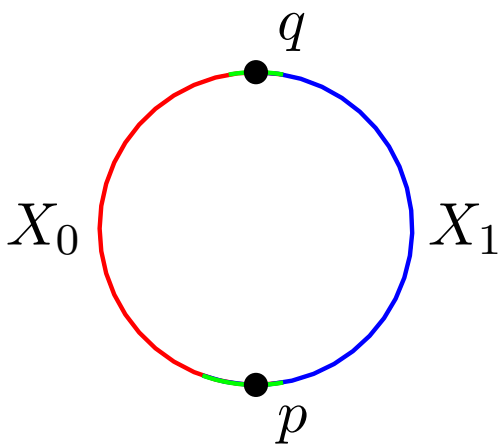
$$\begin{array}{ccc} X_0 \cap X_1 & \xrightarrow{i_0} & X_0 \\ {\scriptstyle i_1}\downarrow & & \downarrow{\scriptstyle j_0} \\ X_1 & \xrightarrow[j_1]{} & X \end{array} \qquad \begin{array}{ccc} \Pi(X_0 \cap X_1) & \xrightarrow{\Pi(i_0)} & \Pi(X_0) \\ {\scriptstyle \Pi(i_1)}\downarrow & & \downarrow{\scriptstyle \Pi(j_0)} \\ \Pi(X_1) & \xrightarrow[\Pi(j_1)]{} & \Pi(X) \end{array}$$

The Seifert-Van Kampen Theorem

Without the fundamental groupoid (using only the fundamental group), we would need to assume that X_0 and X_1 are path-connected in order to have a Van Kampen theorem.

Example: Computing $\pi_1(S^1)$

If we want to split our circle, it would look something like this, with $A = \{p, q\}$.



The intersection is not connected! We need groupoids to apply the Van Kampen theorem.

Example: Computing $\pi_1(S^1)$

The computation now involves using only basic properties of fundamental groupoids and universal properties of the pushout.

1. $\Pi(X_0, A)$ and $\Pi(X_1, A)$ each has exactly one isomorphism
2. $\Pi(X_0 \cap X_1, A)$ has only the identity arrow.
3. $\Pi(S^1, A)$ then has two arrows $u, v : p \to q$ (Van Kampen theorem)
4. $\pi(S^1, p) \cong \{(v^{-1}u)^n : n \in \mathbb{Z}\} \cong \mathbb{Z}$.